



Geospatial Assessment of National and Urban Road Traffic Safety Patterns in Libya

Ibrahim Badi^{1*}, George Baryannis²

¹ Department of Mechanical Engineering, Libyan Academy-Misrata, 2949 Misrata, Libya

² School of Computing and Engineering, University of Huddersfield, HD1 3DH Huddersfield, United Kingdom

* Correspondence: Ibrahim Badi (i.badi@lam.edu.ly)

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Abstract: Road-safety management in data-constrained settings is complicated by the uneven temporal and geographic availability of official records. This study conducts a two-scale Geographic Information System (GIS)-based analysis of road traffic safety in Libya by combining a detailed nationally disaggregated historical baseline with recent national statistics and a longitudinal comparison of Tripoli. Official annual reports for 2016 and 2017 were used to examine national change, spatial divergence among reporting jurisdictions, spatial concentration, and broad geographic patterns of crash frequency and severity. More recent Ministry of Interior statistics provide a national benchmark for 2024 and the first half of 2025, while complete 2025 statistics released by the Tripoli Security Directorate permit a descriptive comparison of the capital across widely separated years. Nationally, recorded crashes fell by 7.0% and recorded deaths by 14.7% between 2016 and 2017, although several jurisdictions moved in the opposite direction. The concentration of recorded crashes and deaths across jurisdictions increased, and the pooled severe-crash share displayed significant positive spatial autocorrelation. The recent national benchmark reported 9,916 recorded crashes and 2,441 deaths in 2024; the first two quarters of 2025 together account for 4,952 crashes and 1,279 deaths. In Tripoli, total recorded crashes increased from 657 in 2016 and 631 in 2017 to 1,628 in 2025, whereas fatal crashes and deaths remained close to their historical levels. Consequently, the fatal-crash share fell from 37.3% and 34.5% to 13.8%. The decline in fatal-crash share may partly reflect changes in reporting practices, traffic exposure, classification or jurisdictional boundaries. The findings support a tiered approach to road-safety assessment in which national spatial screening is combined with current urban monitoring and more consistent geocoded crash-data systems.

Keywords: Road traffic crashes; Geographic Information System; Urban safety; Spatial analysis; Tripoli; Libya; Road-safety management

1 Introduction

Road traffic injury remains a major and persistent public-safety problem. The World Health Organization (WHO) estimated approximately 1.19 million road traffic deaths worldwide in its latest global status report, with the burden concentrated disproportionately in low- and middle-income countries [1]. This pattern is closely connected to rapid motorization, uneven infrastructure quality, weak protection for vulnerable road users, inconsistent enforcement, and limited post-crash capacity. The policy response has increasingly shifted toward a safe system perspective, in which the transport system is expected to anticipate human error and prevent that error from producing fatal or life-changing consequences.

Libya continues to face a severe road-safety burden. The WHO country profile reported 2,218 road traffic fatalities for 2021, while the corresponding WHO estimate was 2,287 fatalities, equivalent to 34.0 deaths per 100,000 population [2]. These values place road safety among the country's important urban and infrastructure-management concerns. Libya's transport system is heavily dependent on private road travel, public transport remains limited, and much of the population and economic activity is concentrated along the northern coastal corridor [3]. In such a setting, traffic safety involves road design, urban growth, infrastructure maintenance, emergency response, and the capacity of institutions to identify where intervention is most urgent.

National totals are necessary for measuring the overall burden, but they can conceal substantial geographic variation. A fall in the national number of crashes may coexist with increases in particular cities. A city with fewer

recorded crashes may also have a higher share of fatal or serious outcomes than a larger city with a higher total count. These distinctions have direct planning implications because the appropriate response to frequent low-severity crashes is not necessarily the same as the response to a smaller number of highly severe crashes. Spatially differentiated evidence is therefore essential when limited road-safety resources must be prioritized across cities and corridors.

Geographic Information System (GIS) provides a practical framework for revealing such differences. Early work demonstrated the value of kernel density estimation (KDE) for identifying spatial concentrations of injury crashes [4], while subsequent studies demonstrated that spatial methods can support the screening of hazardous road sections [5]. Recent applications have extended the approach through hotspot statistics, network-constrained density estimation, and empirical-Bayes procedures [6–9]. GIS has also been combined with multi-criteria methods to translate several dimensions of crash risk into planning-oriented spatial outputs [10]. A recent study further showed how geospatial indicators can be used to identify road exposure and support targeted management in data-constrained transport corridors [11].

The Libyan literature has addressed several parts of the road-safety problem. Research in Tripoli has identified speeding, non-compliance with traffic rules, inadequate lighting, and vehicle condition among the important factors associated with crashes [12]. Other studies have examined comparative safety performance across Libyan cities [13] and the influence of cultural and behavioural factors on mitigation strategies [14]. Broader road-safety research has also investigated intelligent transport technologies for proactive safety management [15] and methodological approaches for analysing crash injury severity [16]. More recent Libyan work has explored intervention selection at a hazardous urban junction when detailed observed data are scarce [17]. Collectively, these studies show that road safety in Libya is shaped by interacting behavioural, infrastructural, technological, and institutional factors. They also demonstrate the continuing need for decision tools that are workable under limited data availability.

A central difficulty is that detailed national traffic records are not consistently available in the same format over time. The 2016 and 2017 official annual reports contain extensive city-level and reporting-jurisdiction tables that can support national spatial analysis [18, 19]. Recent releases provide important national totals for 2024 and 2025, but they do not reproduce the same city-by-city structure [20, 21]. Conversely, the Tripoli Security Directorate has released a complete set of crash-category totals for 2025, enabling an updated urban comparison for the capital [22].

This study therefore adopts a two-scale design. At the national scale, the detailed 2016–2017 reports are used to reconstruct the geographic pattern of recorded crash frequency and severity and to identify local divergence within an apparent national decline in recorded outcomes. Recent national statistics for 2024 and the first half of 2025 are then used to establish whether road-safety burden remains a current policy concern. At the urban scale, Tripoli is examined longitudinally using comparable crash categories for 2016, 2017, and 2025. The analysis addresses three questions: (1) how were crash frequency and severity distributed spatially across Libya during the nationally comparable historical baseline; (2) what do recent national statistics indicate about the persistence of the road-safety burden; and (3) how has the recorded crash profile of Tripoli changed between the historical baseline and 2025?

The study contributes an integrated descriptive assessment of fragmented Libyan administrative records, distinguishing geographic variation in recorded crash frequency and severity from changes in national burden and Tripoli's crash composition. This provides a historical basis for identifying jurisdictions for updated assessment and specifies the data requirements for future road-safety monitoring.

The remainder of this manuscript is structured as follows. Section 2 reviews relevant literature on spatial road-safety analysis and the Libyan context. Section 3 describes the data sources and methodology, while Section 4 presents the historical spatial findings and recent national and Tripoli comparisons. Sections 5 and 6 discuss the findings and their policy implications, respectively. Section 7 outlines the study's limitations, and Section 8 concludes.

2 Literature Review

2.1 Spatial Road-Safety Analysis and Geographic Information System

Spatial road-safety analysis has moved beyond simple crash-location maps toward methods that attempt to identify statistically meaningful concentrations and operationally useful priority areas. KDE remains common because it converts discrete observations into a continuous surface and is easy to interpret alongside road networks and urban land use. Anderson [4] used KDE and K-means clustering to profile injury-crash hotspots in London. Yu et al. [5] compared spatial and conventional hotspot-identification methods and evaluated the applicability of KDE and empirical-Bayes approaches for hazardous road-section screening.

The fact that road crashes occur on connected networks has encouraged network-based approaches. Bisht and Tiwari [7] compared ordinary kriging, planar KDE, and network KDE on an intercity expressway and showed the operational value of network-constrained hotspot identification. Jima and Sipos [8] demonstrated that the selected density method and neighborhood radius can affect the identification of intersection-zone blackspots. Katicha and Flintsch [9] proposed a kernel density empirical Bayes (KDEB) approach to reduce sensitivity to random fluctuation and regression to the mean. These developments are important where precise event-level coordinates and traffic exposure are available.

Citywide and national studies often face a different problem: spatial coverage may be broad while event-level detail is incomplete. Berhanu et al. [6] used GIS and spatial statistics to identify crash hotspots in Addis Ababa, showing how citywide screening can support targeted safety action. Bilaşco and Man [10] combined GIS with the analytic hierarchy process to develop a cumulative road-traffic risk assessment for Romania. Abiodun et al. [11] used geospatial processing and KDE to develop a road exposure index for Nigerian corridors, illustrating the relevance of GIS-based transport risk assessment to urban and regional management in African settings.

The methodological implication is that the spatial technique should match the scale and precision of the source data. Event-level crash points can support road-segment or intersection blackspot analysis. Aggregated reporting-centre totals cannot support that degree of locational inference. They can, however, support broad regional screening when the mapped unit is described correctly and the limits of interpretation are explicit.

The studies most directly relevant to the spatial component of this research are summarized in Table 1. The table emphasizes the relationship between data resolution, analytical method, and how the research is related to the present study.

Table 1. Selected GIS and spatial road-safety studies

Study	Setting	Data and Method	Relevance to This Study
Anderson [4]	London, United Kingdom	Injury crash points; KDE and K-means	Established interpretable spatial hotspot profiling.
Yu et al. [5]	China	Hazardous road-section data; spatial and conventional screening methods	Compared the performance of hotspot-identification approaches.
Berhanu et al. [6]	Addis Ababa, Ethiopia	Urban crash records; GIS hotspot and spatial statistics	Demonstrated citywide spatial screening for targeted safety action.
Bisht and Tiwari [7]	Intercity expressway, India	Fatal crashes; kriging, planar KDE, and network KDE	Showed the value of network-constrained analysis for operational road sections.
Jima and Sipos [8]	Budapest, Hungary	Intersection-zone crashes; point density and KDE	Showed sensitivity of blackspot identification to method and radius.
Katicha and Flintsch [9]	United States	Road-section crash counts; KDEB	Addressed random variation and regression-to-the-mean concerns.
Bilaşco and Man [10]	Romania	National spatial indicators; GIS and AHP	Integrated multiple dimensions into a cumulative spatial risk assessment.
Abiodun et al. [11]	Nigeria	Road corridors; GIS, KDE, and exposure indexing	Connected geospatial risk assessment with transport-management priorities.

Note: GIS = Geographic Information System; KDE = kernel density estimation; KDEB = kernel density empirical Bayes; AHP = analytic hierarchy process.

2.2 Frequency, Severity, and Temporal Benchmarking

Crash frequency is the most direct administrative indicator because it reflects the number of events that police and emergency services record. It is also strongly affected by exposure. Large cities, major corridors, and heavily travelled roads can produce high counts even if the risk per trip is not unusually high. Without population, vehicle-kilometres travelled, traffic flow, or registered-vehicle denominators, counts should therefore be interpreted as recorded burden rather than exposure-adjusted risk.

Severity provides a complementary perspective. Fatal-crash counts, deaths, serious injuries, and the share of crashes producing severe outcomes help distinguish places where the consequences of crashes are unusually harmful. This distinction is important in Libya because previous work has highlighted behavioural risk factors and broader urban safety conditions, including issues related to infrastructure and traffic management [12–14]. A jurisdiction can consequently require intervention because it records many crashes, because its crashes are severe, or because both conditions occur together.

Temporal comparison raises a further challenge. A simple before-and-after percentage may be misleading if reporting systems change, geographic coverage is revised, or nonfatal crashes become more likely to be recorded. Long gaps between observations intensify this concern.

2.3 Road-Safety Research Relevant to Libya and the Remaining Gap

Road-safety research in Libya and related methodological fields has developed along several complementary lines. Elturki and Ali [12] investigated factors affecting crashes in Tripoli and reported strong concern about speeding,

non-compliance with road rules, road lighting, and vehicle condition. Elmansouri et al. [3] placed traffic safety within the broader structure of Libya's urban transport system, emphasizing the relationship between infrastructure, mobility, and institutional planning. Geurts and Wets [15] demonstrated how clustering techniques can be applied to identify accident-prone locations and support spatial prioritization in road-safety analysis. Savolainen et al. [16] reviewed statistical approaches for identifying factors associated with crash injury severity and highlighted the importance of methodological choices in road-safety analysis.

At the city-comparison level, previous research developed integrated approaches for evaluating and ranking traffic-safety conditions across Libyan cities [13]. Subsequent studies incorporated cultural norms and behavioural risks into the prioritization of mitigation strategies [14]. More recent intersection-level research has moved toward finer urban-scale analysis by combining microsimulation and decision analysis to evaluate low-cost safety measures under sparse data conditions [17]. These studies provide useful decision frameworks, but they do not resolve the question addressed here: whether national changes in administrative crash totals are geographically uniform and how an older spatial baseline can be connected responsibly with recent evidence.

Table 2 positions the present analysis within the broader body of transport and road-safety research relevant to Libya. The reviewed studies provide valuable insights into specific aspects of road safety, but limited attention has been given to combining a national spatial assessment of crash frequency, severity, and concentration with recent national indicators and an updated urban comparison. The present study addresses this combination using the available administrative records.

Table 2. Selected transport and road-safety studies supporting the Libyan case analysis

Study	Focus	Approach	Contribution to the Present Study
Elmansouri et al. [3]	Urban transportation in Libya	National transport-system review	Provides the broader mobility and infrastructure context.
Elturki and Ali [12]	Crash factors in Tripoli	Questionnaire and RII	Identifies behavioural, road, vehicle, and environmental factors.
Badi et al. [13]	Traffic safety across Libyan cities	Integrated city-ranking model	Supports comparison of spatially distinct urban safety conditions.
Badi et al. [14]	Cultural and behavioural influences	Hybrid MCDM	Connects behavior and institutional measures with mitigation priorities.
Geurts and Wets [15]	Accident-prone location identification	Clustering techniques for spatial safety analysis	Supports the identification and prioritization of hazardous locations using spatial methods.
Savolainen et al. [16]	Crash injury severity analysis and risk factors	Statistical modelling review	Provides methodological foundations for identifying factors associated with road-safety outcomes.
Badi et al. [17]	Hazardous urban intersection	Microsimulation and decision analysis	Shows site-level intervention planning in a data-sparse Libyan setting.

Note: RII = relative importance index; MCDM = multi-criteria decision-making.

3 Methodology

3.1 Study Design

The study was designed as a two-scale GIS-based retrospective and comparative analysis. The first scale is national and spatial. It uses the Statistical Books for 2016 and 2017 published by the Bureau of Statistics and Census, Libya, which provide national-level statistical records for GIS-based analysis [18, 19]. These reports contain national totals and tables by town or traffic reporting jurisdiction, including fatal crashes, serious-injury crashes, minor-injury crashes, damage-only crashes, recorded deaths, injured persons, month, time of day, and road environment. Because the reporting structure is highly detailed and broadly comparable across the two years, these reports form the historical spatial baseline.

The second scale is contemporary and partly non-spatial. A Ministry of Interior statement reported 9,916 traffic crashes, 2,441 deaths, and 4,139 serious injuries nationally in 2024, together with first-quarter 2025 totals [20]. A subsequent release by the Traffic Affairs and Licensing Department reported second-quarter 2025 totals [21]. These releases are used to establish a recent national benchmark but are not mapped by city because the published data are not geographically disaggregated. Separately, statistics released by the Tripoli Security Directorate for the full 2025 calendar year provide fatal, serious-injury, minor-injury, and damage-only crash totals for the capital's reporting jurisdiction [22]. These data are compared with the Tripoli rows in the 2016 and 2017 national reports.

The historical reports support spatial comparison, while the recent national releases support current burden assessment. Tripoli is the only location used for the temporal city comparison because the 2025 release provides a complete annual set of categories that can be aligned with the earlier reports.

Table 3 identifies each source, its geographic resolution, and its role in the analysis.

Table 3. Data sources, coverage, variables, and analytical roles

Source	Coverage	Main Variables Used	Analytical Role
Statistical Book 2016 [18]	National; city-level and reporting-jurisdiction tables	Crashes by severity, deaths, injuries, month, road environment	Historical national baseline and GIS mapping
Statistical Book 2017 [19]	National; city-level and reporting-jurisdiction tables	Crashes by severity, deaths, injuries, month, road environment	Paired historical comparison and GIS mapping
Ministry of Interior statement [20]	National; 2024 annual data and Q1 2025 data	Total crashes, deaths, serious injuries; Q1 minor injuries	Recent national benchmark
Traffic Affairs and Licensing statistics [21]	National; Q2 2025 data	Total crashes, deaths, serious injuries, minor injuries	Provides the national first-half 2025 benchmark
Tripoli Security Directorate statistics [22]	Tripoli reporting jurisdiction; full-year 2025	Fatal, serious-injury, minor-injury, and damage-only crashes; persons affected	Historical urban comparison with 2016 and 2017

Note: Q1 = first quarter; Q2 = second quarter; GIS = Geographic Information System.

3.2 Data Harmonization and Indicators

The historical tables were transcribed into a structured dataset and checked against national totals printed in the annual reports. City names were harmonized across Arabic and English spellings, and only reporting-centres with unambiguous geographic locations were retained for spatial analysis. The mapped subset contains 38 reporting-centres and represents 92.79% of the matched historical total crash records. Reporting-centres with uncertain geographic definitions or inconsistent labels were retained in national totals where appropriate but excluded from the GIS layer.

The main indicators were total recorded crashes, fatal-crash records, serious-injury crash records, minor-injury crash records, damage-only crash records, recorded deaths, and numbers of persons reported with serious or minor injuries. Relative change was calculated as the difference between the later and earlier value divided by the earlier value, expressed as a percentage. A severe-crash share was calculated for each mapped reporting-centre by combining fatal-crash and serious-injury crash records and dividing by total recorded crashes.

For Tripoli, two descriptive severity indicators were added. The fatal-crash share expresses fatal-crash records as a percentage of all recorded crashes. Deaths per 100 recorded crashes divides the number of recorded deaths by total crash records and multiplies the result by 100. These ratios help describe the changing composition of the administrative record.

The harmonized variables and their interpretation are summarized in Table 4.

Table 4. Analytical indicators and interpretation

Indicator	Operational Meaning	Interpretation
Total recorded crashes	All crash records reported for a jurisdiction or national period	Administrative crash burden
Fatal-crash records	Crash records classified as involving a fatal outcome	Frequency of fatal crash events
Recorded deaths	Persons reported killed in recorded crashes	Human fatality burden
Serious-injury crash records	Crashes classified as involving serious injury	Frequency of serious-injury events
Severe-crash share	Fatal plus serious-injury crash records divided by all crash records	Relative severity composition
Relative change	Percentage change between two observations	Direction and magnitude of change
Fatal-crash share	Fatal-crash records divided by total crash records	Tripoli crash-composition indicator
Deaths per 100 recorded crashes	Recorded deaths relative to total crash records	Tripoli consequence indicator

3.3 Statistical Analysis

National descriptive analysis compared 2016 and 2017 totals and calculated absolute and percentage changes. City-level rank stability was assessed with Spearman's rank correlation because the distribution of jurisdictional crash counts was strongly skewed. The Gini coefficient was used as a descriptive measure of concentration to examine how total crashes and recorded deaths were distributed across reporting jurisdictions. In this study, the Gini coefficient reflects the concentration of road-traffic burden across jurisdictions rather than socioeconomic inequality. A larger Gini coefficient indicates that a greater share of the recorded burden is concentrated in a smaller number of jurisdictions.

The monthly distributions for 2016 and 2017 were compared descriptively and using Spearman correlation, to assess whether the ranking of months remained similar between years. For the recent benchmark, the published 2024 totals were compared with the historical national values. First-half 2025 totals were obtained by summing the first- and second-quarter releases. No annualization of the six-month totals was performed because the purpose was to report the observed first-half burden rather than estimate annual totals. For Tripoli, the 2025 crash categories were aligned with the 2016 and 2017 reporting categories and percentage changes were computed.

3.4 Geographic Information System Preparation and Spatial Analysis

Each retained reporting-centre was represented by a geographic point corresponding to the relevant reporting-centre location. Counts were attached to these points for the two historical years. The resulting maps are reporting-centre weighted surfaces.

Four spatial products were prepared. The first shows smoothed average crash intensity based on reporting-centre weighted observations across the two historical years. The second shows smoothed average reporting-centre weighted death intensity. The third maps the pooled severe-crash share, which separates locations where the composition of recorded crashes was more severe from locations where high frequency was driven mainly by less severe categories. The fourth combines median average crash frequency and median pooled severe-crash share to classify reporting-centres into four management groups: high frequency-high severity, high frequency-lower severity, lower frequency-high severity, and lower frequency-lower severity. These groups are defined relative to the sample medians across the 38 mapped reporting-centres.

The smoothed surfaces use a broad bandwidth selected for regional-scale screening rather than fine-scale hotspot detection. Global Moran's I was calculated for average crash frequency, average recorded deaths, pooled severe-crash share, and absolute crash-count change using a symmetric four-nearest-neighbor spatial-weights structure. A four-nearest-neighbor specification was selected to provide sufficient spatial connectivity among the retained reporting-centres while preserving local spatial relationships. Statistical significance was evaluated with 999 random permutations following the logic of spatial autocorrelation testing introduced by Moran [23]. The permutation test assessed whether similar values were more spatially clustered than expected under random rearrangement.

3.5 Data Quality, Ethics, and Reproducibility

All sources are aggregated administrative statistics or publicly released institutional summaries. No individual identifiers, medical records, or personal data were used. The historical national values were checked against totals printed in the official reports. Recent national statistics were taken from the Libyan News Agency reports of Ministry of Interior and Traffic Affairs and Licensing statements [20, 21]. The 2025 Tripoli values were reported as statistics released by the Tripoli Security Directorate and were cross-checked against an independent English-language report reproducing the same figures [22].

The analysis preserves the distinction between crash records and persons. A fatal crash is an event count, whereas recorded deaths are person counts. The same distinction applies to injury crash categories and injured persons. Maintaining this separation is essential because the number of people affected can exceed the number of crash records.

4 Results

4.1 Historical National Baseline

The paired national reports show a reduction in most severe outcomes between 2016 and 2017. Total recorded crashes decreased from 4,994 to 4,644, a fall of 7.0%. Fatal-crash records declined from 1,945 to 1,623, and recorded deaths decreased from 2,414 to 2,059. Serious-injury crash records fell modestly, from 1,017 to 975, while damage-only records increased slightly from 1,475 to 1,494. The calculated severe-crash share decreased from 59.35% in 2016 to 55.90% in 2017 [18, 19].

Table 5 presents the national comparison. The broad signal is a reduction in recorded fatal outcomes, but the reduction was not shared evenly across crash categories. Damage-only records increased despite the decline in total crashes, while fatal-crash records fell more quickly than total volume. This early shift in composition becomes important when the older national results are considered alongside the much later Tripoli data.

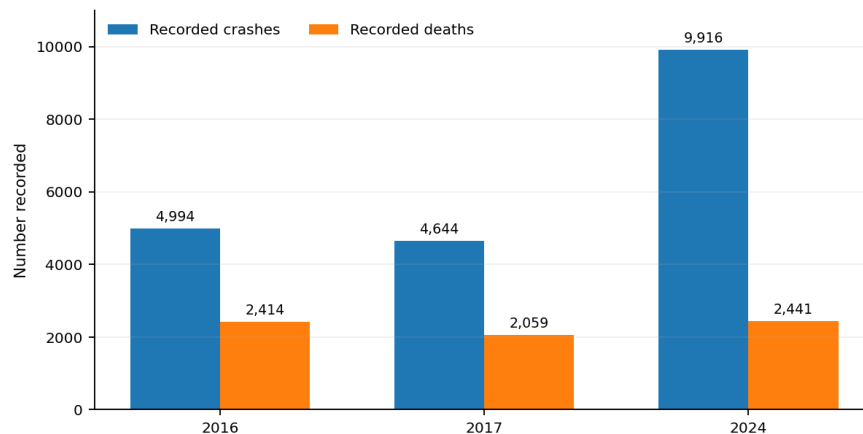
Table 5. National road-traffic indicators in the paired historical reports

Indicator	2016	2017	Absolute Change	Relative Change
Total recorded crashes	4,994	4,644	−350	−7.0%
Fatal-crash records	1,945	1,623	−322	−16.6%
Serious-injury crash records	1,017	975	−42	−4.1%
Minor-injury crash records	557	552	−5	−0.9%
Damage-only crash records	1,475	1,494	+19	+1.3%
Recorded deaths	2,414	2,059	−355	−14.7%
Persons with serious injuries	2,156	2,015	−141	−6.5%
Persons with minor injuries	1,540	1,398	−142	−9.2%
Calculated severe-crash share	59.35%	55.90%	−3.45 percentage points	−5.8%

4.2 Contemporary National Benchmark

Recent Ministry of Interior statistics show a substantial national road-safety burden. The 2024 statement reported 9,916 crashes, 2,441 deaths, and 4,139 serious injuries nationally [20]. Relative to the 2017 administrative total, the number of recorded crashes is 113.5% higher, while recorded deaths are 18.6% higher. The very different rates of change can be attributed to a range of underlying mechanisms: increased traffic exposure, population change, reporting coverage, enforcement, classification practice, and changes in the mix of crash severities.

Figure 1 places the 2024 national benchmark beside the two historical reports. The contrast between crash volume and deaths is notable: the recent crash count is much larger, whereas the death count is closer to the historical range.

**Figure 1.** National recorded crash and death totals in the historical reports and the 2024 benchmark

The first quarter of 2025 recorded 2,282 crashes, 660 deaths, 1,006 serious injuries, and 656 minor injuries [20]. The second quarter recorded 2,670 crashes, 619 deaths, 990 serious injuries, and 777 minor injuries [21]. In the first half of 2025, the national totals were 4,952 recorded crashes, 1,279 deaths, 1,996 serious injuries, and 1,433 minor injuries.

Table 6 provides the recent national benchmark in numerical form. As shown in Figure 2, recorded crashes increased from the first to the second quarter, while deaths and serious injuries were slightly lower. Minor injuries were higher in the second quarter.

Table 6. Recent national road-traffic benchmark

Period	Recorded Crashes	Deaths	Serious Injuries	Minor Injuries
2024 full year	9,916	2,441	4,139	Not reported in source
2025 Q1	2,282	660	1,006	656
2025 Q2	2,670	619	990	777
2025 H1, sum of Q1 and Q2	4,952	1,279	1,996	1,433

Note: Q1 = first quarter; Q2 = second quarter; H1 = first half of the year. The 2025 H1 values are calculated as the sum of Q1 and Q2 official releases and do not represent a separately published annual statistic.

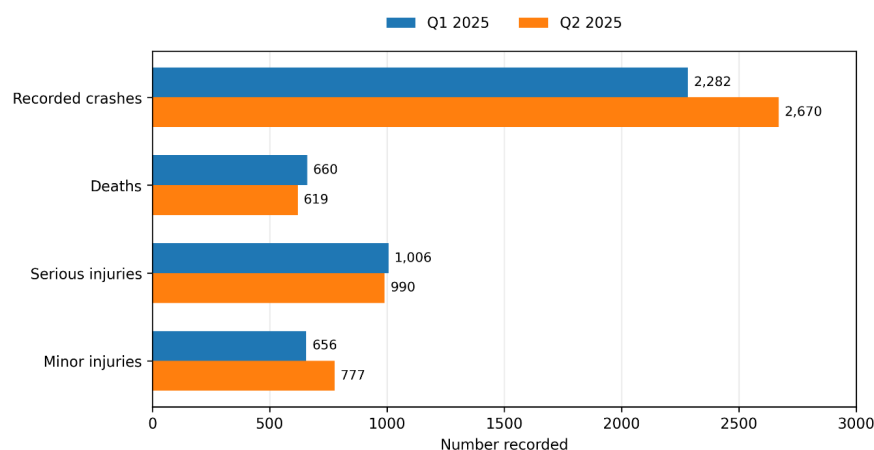


Figure 2. National road-traffic outcomes reported for the first and second quarters of 2025

Note: Q1 = first quarter; Q2 = second quarter.

4.3 Local Divergence Within the Historical National Baseline

The national decline concealed large differences among reporting jurisdictions. Ajdabiya increased from 214 to 359 recorded crashes, a rise of 67.8%, while Misrata increased from 311 to 434 crashes, or 39.5%. Benghazi remained the largest reporting jurisdiction by crash count but declined from 1,097 to 1,011 crashes. Tripoli declined from 657 to 631. Several western and eastern jurisdictions recorded substantially larger percentage reductions in recorded crashes, including Al Wahat, Ras al Hilal, Al Zawiya, Al Jafara, Tobruk, and Shahat [18, 19].

Changes in deaths did not mirror changes in crash volume. Misrata's recorded deaths rose from 198 to 271, while Ajdabiya's fell from 84 to 42 despite its large increase in total crash records. Benghazi's deaths increased from 150 to 160 while its total crash count declined. These contrasts reinforce the need to preserve separate frequency and severity measures.

Table 7 reports selected jurisdictions that capture the main forms of divergence. The strong increase in Ajdabiya and Misrata is particularly important because the national total was falling at the same time.

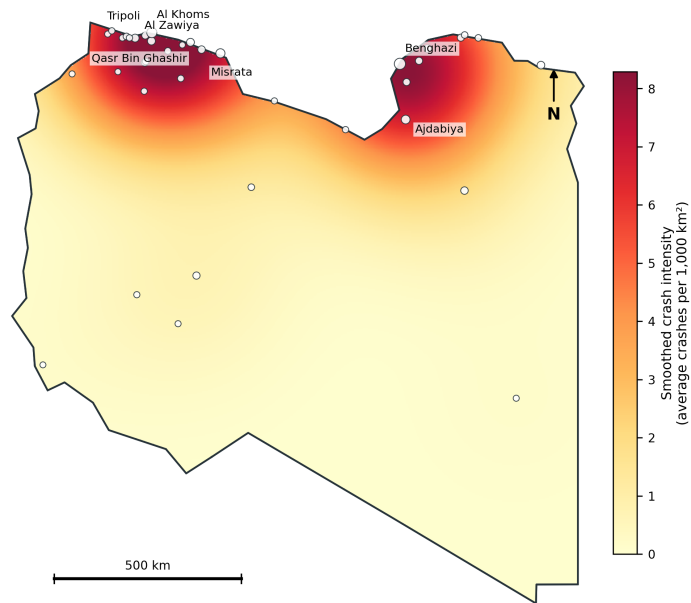
Table 7. Selected reporting-jurisdiction changes in the historical baseline

Jurisdiction	Crashes 2016	Crashes 2017	Crashes Change	Crashes Change (%)	Deaths 2016	Deaths 2017	Death Change	Death Change (%)
Ajdabiya	214	359	+145	+67.8%	84	42	-42	-50.0%
Misrata	311	434	+123	+39.5%	198	271	+73	+36.9%
Benghazi	1,097	1,011	-86	-7.8%	150	160	+10	+6.7%
Tripoli	657	631	-26	-4.0%	278	241	-37	-13.3%
Al Wahat	139	80	-59	-42.4%	45	24	-21	-46.7%
Ras al Hilal	92	56	-36	-39.1%	58	39	-19	-32.8%
Al Zawiya	191	117	-74	-38.7%	100	81	-19	-19.0%
Al Jafara	123	82	-41	-33.3%	126	76	-50	-39.7%
Tobruk	157	105	-52	-33.1%	82	56	-26	-31.7%
Shahat	50	34	-16	-32.0%	16	2	-14	-87.5%

Note: Percentage changes were calculated relative to the 2016 values.

4.4 Spatial Distribution of Historical Frequency and Severity

The geographic distribution of the paired historical data is shown in Figures 3–6. Figure 3 maps the reporting-centre weighted distribution of average recorded crashes across the geocoded reporting-centres. The resulting surfaces represent reporting-centre weighted distributions rather than continuous estimates of crash occurrence risk. The strongest concentrations are located along the northern coastal belt, with a prominent western concentration around the Tripoli metropolitan area and a separate eastern concentration around Benghazi. Misrata and Ajdabiya appear as important intermediate centres. The pattern reflects the concentration of population, mobility, and reporting activity along the coastal urban system.

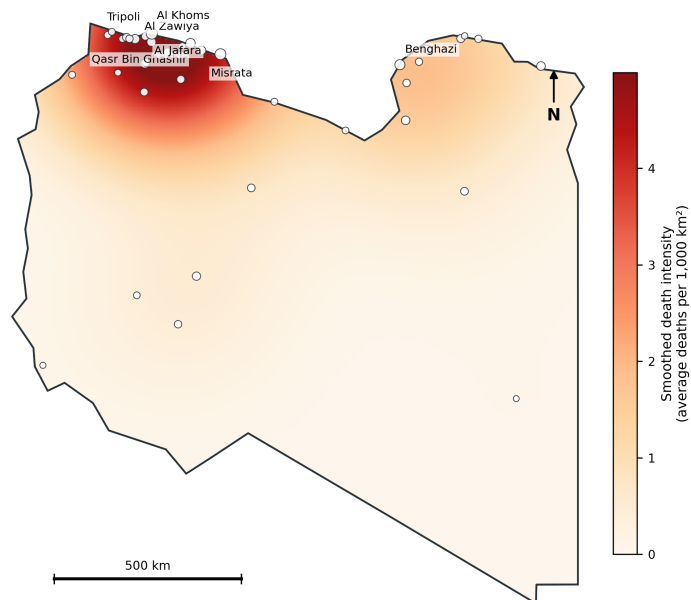


Reporting-centre weighted surface; locations are not individual crash points.

Figure 3. Reporting-centre weighted surface of average recorded crash concentration across the paired historical reports

Note: The displayed surface represents a smoothed reporting-centre weighted distribution and does not represent population-normalized crash rates or individual crash locations.

Figure 4 maps the reporting-centre weighted distribution of average recorded deaths across the geocoded reporting-centres. Its surface is less dominated by the highest crash-count centre than the frequency map, showing that the geography of fatal outcomes is not identical to the geography of total crash volume. The distinction is particularly important for management because a reporting jurisdiction can remain a priority even when it does not rank among the highest by total events.



Reporting-centre weighted surface; locations are not individual crash points.

Figure 4. Reporting-centre weighted surface of average recorded-death intensity across the paired historical reports

Note: The displayed surface represents a smoothed reporting-centre weighted distribution and does not represent population-normalized death rates or individual crash locations.

Figure 5 shifts the focus from counts to composition by mapping the pooled share of fatal and serious-injury crash records. The broad surface suggests that areas with higher severe-crash shares are not limited to jurisdictions with the largest crash totals. This observation is supported by the spatial-autocorrelation results. Global Moran's I was 0.040 for average crash frequency and 0.062 for average recorded deaths, neither statistically significant. In contrast, the pooled severe-crash share calculated from the combined 2016–2017 records produced Moran's I = 0.257 (permutation $p = 0.005$), indicating significant positive spatial association.

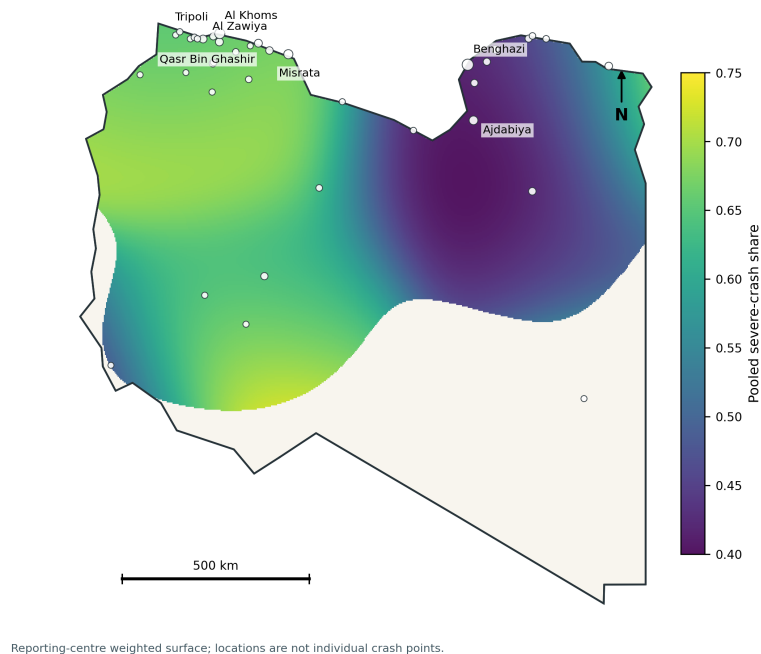


Figure 5. Reporting-center weighted surface of pooled severe-crash share across the paired historical reports

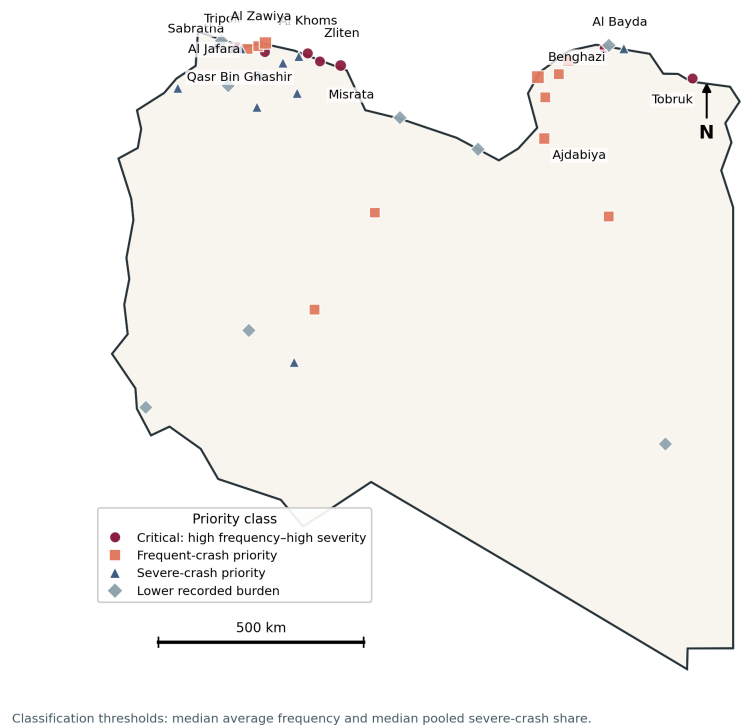


Figure 6. Historical frequency-severity road-safety priority classes for the geocoded reporting-centres

The frequency-severity classification in Figure 6 translates these two dimensions into a planning-oriented screening

tool. Eight reporting-centers fall in the high-frequency and high-severity group: Al Bayda, Al Jafara, Al Khoms, Misrata, Qasr Bin Ghashir, Sabratha, Tobruk, and Zliten. Eleven centers, including Tripoli, Benghazi, Ajdabiya, Sabha, and Al Zawiya, fall in the high-frequency but lower-severity group. A further eleven centers have lower frequency but higher severity. This separation prevents high-volume centers from automatically absorbing all attention when other locations show a more severe crash composition.

Table 8 summarizes the main statistical diagnostics supporting the historical spatial interpretation. The Spearman's ρ of 0.899 indicates that the relative ordering of jurisdictions by total crash count was highly stable even though individual locations changed substantially. The Gini coefficient increased from 0.671 to 0.705 for total crashes and from 0.561 to 0.609 for recorded deaths, showing that the burden became more concentrated across jurisdictions. The correlation between the monthly profiles was not statistically significant (Spearman's $\rho = -0.406$, $p = 0.191$), providing insufficient evidence of a consistent ranking of months between the two years.

Table 8. Statistical diagnostics for the historical national analysis

Analysis	Statistic	Value	<i>p</i> -Value	Interpretation
Reporting-jurisdiction total-crash rank stability	Spearman's ρ	0.899	<0.001	Strong stability in jurisdiction ordering
Monthly-profile association	Spearman's ρ	-0.406	0.191	No statistically significant association
Crash-count concentration, 2016	Gini coefficient	0.671	—	Concentration measured by Gini coefficient
Crash-count concentration, 2017	Gini coefficient	0.705	—	Higher concentration than 2016
Death concentration, 2016	Gini coefficient	0.561	—	Concentration measured by Gini coefficient
Death concentration, 2017	Gini coefficient	0.609	—	Higher concentration than 2016
Average crash frequency	Moran's I	0.040	0.682	No significant global autocorrelation
Average recorded deaths	Moran's I	0.062	0.545	No significant global autocorrelation
Pooled severe-crash share	Moran's I	0.257	0.005	Significant positive spatial autocorrelation
Absolute crash-count change	Moran's I	0.022	0.818	No significant global autocorrelation

Note: — indicates that a *p*-value is not applicable to the corresponding analysis.

4.5 Long-Term Change in the Tripoli Reporting Jurisdiction

Tripoli provides the strongest available bridge between the detailed historical reports and current city-level evidence. In 2016, the Tripoli row recorded 657 crashes: 245 fatal crashes, 115 serious-injury crashes, 70 minor-injury crashes, and 227 damage-only crashes. These events were associated with 278 deaths, 181 people with serious injuries, and 209 people with minor injuries [18]. In 2017, the corresponding totals were 631 crashes, 218 fatal crashes, 85 serious-injury crashes, 87 minor-injury crashes, and 241 damage-only crashes, with 241 deaths, 158 serious injuries, and 235 minor injuries [19].

The 2025 statistics released by the Tripoli Security Directorate report 225 fatal crashes causing 251 deaths, 375 serious-injury crashes affecting 492 people, 320 minor-injury crashes affecting 624 people, and 708 damage-only crashes [22]. Summing the four event categories gives 1,628 recorded crashes. Figure 7 shows how strongly the composition of the administrative record changed. Total recorded events increased sharply, but the increase was concentrated in serious-injury, minor-injury, and damage-only categories rather than fatal-crash events.

The numerical comparison is presented in Table 9. From 2017 to 2025, total recorded crashes increased by 158.0%. Fatal-crash records increased by only 3.2% and recorded deaths by 4.1%, whereas serious-injury crash records increased by 341.2%, minor-injury crash records by 267.8%, and damage-only records by 193.8%. These differences indicate a change in the composition of recorded events, potentially influenced by changes in reporting coverage, and cannot be attributed to a single safety mechanism.

Figure 8 makes the compositional change explicit. Fatal crashes represented 37.3% of Tripoli's recorded crashes in 2016 and 34.5% in 2017, compared with 13.8% in 2025. Recorded deaths per 100 crashes similarly fell from 42.3 and 38.2 to 15.4. This decline is consistent with a lower recorded fatality-related severity measure, but it can also arise if nonfatal and damage-only crashes are captured more completely in the recent reporting system.

This distinction is important because the absolute number of deaths did not fall sharply over the long interval. The 2025 total of 251 deaths is close to the 241 deaths reported in 2017 and below the 278 reported in 2016. The current burden therefore remains substantial even though fatal events represent a smaller share of a much larger recorded crash total.

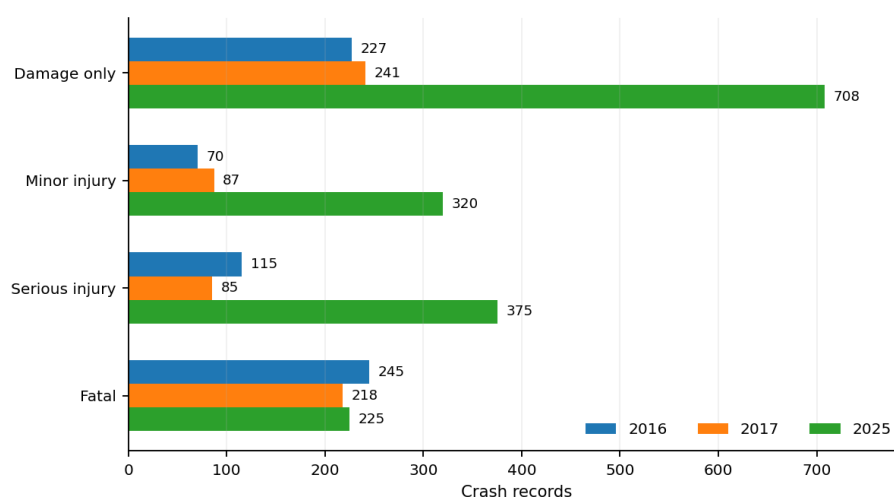


Figure 7. Distribution of recorded crash categories in the Tripoli reporting jurisdiction

Table 9. Long-term comparison of the Tripoli reporting jurisdiction

Indicator	2016	2017	2025	Relative Change 2017–2025
Total recorded crashes	657	631	1,628	+158.0%
Fatal-crash records	245	218	225	+3.2%
Recorded deaths	278	241	251	+4.1%
Serious-injury crash records	115	85	375	+341.2%
Persons with serious injuries	181	158	492	+211.4%
Minor-injury crash records	70	87	320	+267.8%
Persons with minor injuries	209	235	624	+165.5%
Damage-only crash records	227	241	708	+193.8%
Fatal-crash share	37.3%	34.5%	13.8%	–20.7 percentage points
Recorded deaths per 100 recorded crashes	42.3	38.2	15.4	–22.8

Note: Fatal crashes refer to crash events, whereas recorded deaths refer to persons killed in those crashes.

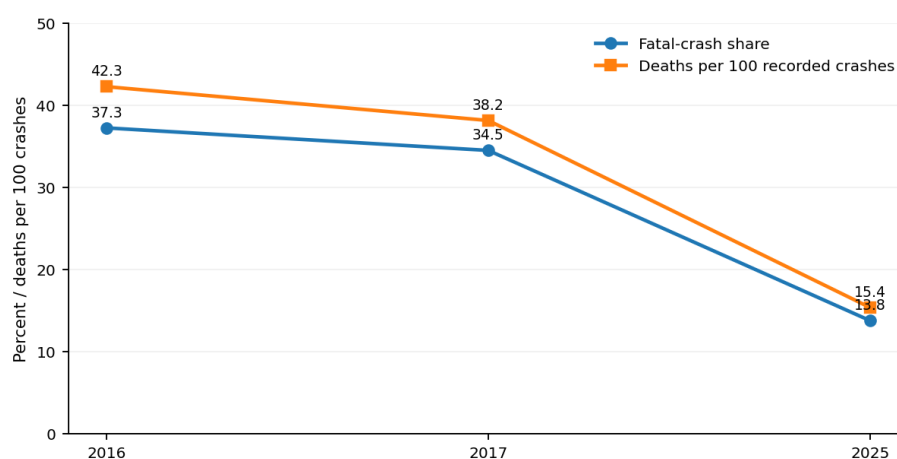


Figure 8. Fatal-crash share and recorded deaths per 100 crashes in the Tripoli reporting jurisdiction

5 Discussion

5.1 Interpreting National Change Across Unequal Data Periods

The historical national comparison shows why aggregate reductions should not be interpreted as spatially uniform changes. Between 2016 and 2017, Libya recorded fewer crashes and fewer fatal outcomes, yet important jurisdictions moved in different directions. Misrata and Ajdabiya increased substantially, while several other centres declined. The

strong Spearman rank correlation indicates that the broad hierarchy of reporting jurisdictions remained stable, but the rising Gini coefficients indicate that the recorded burden became more concentrated across reporting jurisdictions. These results support a planning approach that combines national monitoring with local screening.

The recent national releases change the interpretation of the older data. They show that road safety remains a current challenge. The 2024 reported total of 9,916 crashes is more than twice the 2017 count, while the 2,441 recorded deaths are only moderately higher than the 2017 death total. The first half of 2025 alone accounts for 4,952 crashes and 1,279 deaths. The recent releases establish the continuing national burden, while the historical reports describe its earlier geographic distribution; they do not establish whether that distribution persists.

5.2 Insights From the Tripoli Comparison

Tripoli provides a useful comparison of long-term recorded changes because the 2025 release uses similar broad crash categories as the historical reports. The composition of recorded crashes changed considerably. Fatal-crash counts and deaths in 2025 remain close to the earlier levels, while serious-injury, minor-injury, and damage-only crash records are substantially higher. This produced a marked fall in fatal-crash share and deaths per 100 recorded crashes.

Several interpretations are possible and they are not mutually exclusive. First, the composition of actual crashes may have shifted toward less fatal outcomes because of changes in vehicles, emergency response, driving environments, or the locations where crashes occur. Second, population growth and motorization may have increased the number of lower-severity encounters while fatal events remained relatively stable. Third, reporting completeness may have improved, particularly for injury and damage-only crashes. Fourth, the jurisdiction covered by the Tripoli Security Directorate in 2025 may not correspond perfectly to the traffic reporting unit used in the earlier national reports. Without traffic exposure and a formal metadata history, these mechanisms cannot be separated confidently.

This caution identifies a practical management question: why has the administrative profile of Tripoli changed so strongly? A road-safety authority could investigate this by comparing police reporting rules, hospital linkage, vehicle registrations, road-network expansion, population exposure, and enforcement practices across the periods.

The finding also complements earlier Tripoli research. Elturki and Ali [12] found that respondents gave high importance to speeding, disregard of traffic rules, poor lighting, and vehicle defects. More recent Libyan work has emphasized technology, behavioural interventions, and targeted site treatment [14, 17]. Evaluating these interventions requires consistent administrative records of crash frequency and severity.

5.3 Spatial Concentration and Differentiated Urban Safety Management

The GIS results show that frequency and severity produce related but non-identical spatial patterns. The broad coastal concentration of crash frequency is expected in a country where population and mobility are concentrated in northern urban corridors [3]. However, the significant positive spatial autocorrelation of severe-crash share suggests that severity composition has a more coherent geographic structure than raw event counts. This finding argues against a uniform allocation rule based only on total crashes.

The four-class map identifies different historical crash profiles that can guide updated local assessment. If confirmed by contemporary data, high-frequency and high-severity profiles would justify examining both crash prevention and the reduction of fatal and serious outcomes. High-frequency but lower-severity profiles would suggest examining recurrent traffic conflicts, while lower-frequency but high-severity profiles would suggest investigating the circumstances of severe crashes and access to emergency care. The lower-frequency and lower-severity group should remain under monitoring. Specific interventions require current evidence on crash locations, exposure and contributing factors.

This logic aligns with the progression of GIS road-safety research from visual hotspot detection toward decision-oriented screening [4–11]. The spatial resolution of the maps is constrained by the aggregation of crash records at reporting-center level. Their value lies in identifying candidates for further assessment. Current crash coordinates, traffic exposure and road characteristics are needed before determining investment priorities or selecting street-level interventions.

5.4 Implications for Data Governance and Future Geographic Information System Monitoring

Fragmentation across time and geographic scale limits the use of Libya's road-safety data. Libya has detailed historical annual reports, current national releases, and local statistics from security directorates, yet these sources are not available as one consistent geocoded series. The immediate policy opportunity is therefore institutional: standardize the variables that every traffic department reports, preserve annual definitions, and publish a machine-readable national dataset with event coordinates or road-segment identifiers.

A national road-safety observatory could operate in two layers. The first layer would maintain countrywide indicators by municipality or traffic jurisdiction, allowing annual comparison of crash frequency, deaths, serious injuries, road type, time of day, and major behavioural factors. The second layer would maintain event-level geocoded records for high-priority urban areas. This would allow national screening to be followed by network KDE,

intersection analysis, exposure adjustment, and before-after evaluation of interventions. The structure would also support intelligent transport applications proposed in earlier research [15].

Linkage with hospital and emergency-service data is equally important. Police records are indispensable for location and enforcement information, but serious injuries can be misclassified if medical outcomes are not connected to the crash record. A shared identifier across police, ambulance, and hospital systems would improve the distinction between minor injury, serious injury, and death and would make future severity comparisons more reliable.

6 Policy Implications

Five key policy implications arise from this study. The first is to adopt a geographically differentiated allocation approach. National totals should be used to monitor overall burden, but investment decisions should be informed by city-level frequency and severity. The historical classification identifies candidates for updated assessment, but contemporary jurisdiction-level evidence is needed before it can inform investment priorities.

Second, Tripoli represents an important case for improved data integration. The 2025 statistics show that the number of recorded nonfatal and damage-only crashes has expanded considerably, while fatal crashes remain close to historical levels. This creates an opportunity to determine whether reporting completeness improved and whether specific urban corridors account for the growth. The next analytical step should use geocoded 2025 crash records, traffic volumes, road hierarchy, and intersection geometry to locate the current burden within the capital.

Third, serious injury should receive the same management attention as fatalities. The 2025 national quarterly releases report almost 2,000 serious injuries in the first half of the year, and the Tripoli release reports 492 seriously injured persons for the full year. A safety program that reports only deaths can miss a large burden of life-changing injury.

Fourth, road-safety reporting should be linked directly to infrastructure planning. High-priority jurisdictions should undergo road-safety audits, speed-management review, nighttime visibility assessment, junction and median evaluation, pedestrian-risk assessment, and emergency-response review. Earlier Libyan studies already identify speeding, road condition, lighting, and behavioural compliance as relevant concerns [12, 14]. The historical GIS screening can inform the selection of jurisdictions for preliminary assessment, with audit priorities subsequently determined using current crash records and local evidence.

Finally, annual publication should use stable definitions and geographic identifiers. A short machine-readable table containing jurisdiction code, municipality, crash type, deaths, serious injuries, minor injuries, damage-only crashes, road class, and month would allow the national system to move from occasional descriptive reports to continuous evidence-based management.

7 Limitations

The following limitations should be considered when evaluating the findings of this research. Comparisons across years are constrained by possible changes in reporting completeness, injury classification and administrative coverage, including the boundaries of the Tripoli reporting jurisdiction. Observed differences in crash counts and severity composition may therefore reflect both changes in road safety and changes in administrative recording. Uneven reporting completeness across jurisdictions may also affect spatial comparisons.

The absence of consistent exposure denominators prevents exposure-adjusted risk comparisons. The aggregation of crash records at reporting-center level also limits spatial inference, preventing the identification of hazardous road segments or intersections.

The available contemporary national totals and Tripoli statistics do not establish whether the 2016–2017 spatial patterns persist. The historical classifications consequently require updated assessment before informing current investment priorities. Median-based classes depend on the included sample, while severe-crash shares may be unstable for reporting-centres with small crash counts.

8 Conclusions

This study combines two types of evidence that are often separated in data-constrained road-safety research: a detailed historical spatial baseline and recent but less geographically complete statistics. The paired national reports show that Libya's recorded crash and fatality burden declined between 2016 and 2017, but the recorded reductions were uneven across cities. The spatial analysis identifies strong coastal concentration, increasing concentration in the distribution of crashes and deaths, and a severity pattern that is not identical to the frequency pattern.

Recent official releases confirm that the road-safety burden remains substantial. Libya recorded 9,916 crashes and 2,441 deaths in 2024, while the first two quarters of 2025 together account for 4,952 crashes and 1,279 deaths. Tripoli adds a more detailed urban perspective. Its total recorded crash count rose from 657 in 2016 and 631 in 2017 to 1,628 in 2025, but fatal-crash records and deaths remained close to the historical range. The resulting reduction in fatal-crash share is important, but it cannot be interpreted independently of reporting coverage, traffic exposure, and administrative comparability.

For urban development and management, the findings support geographically differentiated road-safety monitoring and stronger data integration. The historical spatial analysis identifies candidates for updated assessment, while current jurisdiction-level and event-level data are needed to establish present priorities and guide corridor and intersection-level interventions. Standardized definitions, stable geographic identifiers and geocoded records would enable more consistent monitoring and evaluation.

Author Contributions

Conceptualization, I.B. and G.B.; methodology, I.B.; software, I.B.; validation, G.B.; formal analysis, I.B.; investigation, G.B.; writing—original draft preparation, I.B.; writing—review and editing, G.B.; visualization, I.B. and G.B.; project administration, G.B. All authors have read and agreed to the published version of the manuscript.

Data Availability

The historical data used in the national analysis are derived from the official Statistical Book cited in this article. Recent national statistics were obtained from the cited Libyan News Agency releases. The 2025 Tripoli statistics were obtained from the cited public statement released by the Tripoli Security Directorate.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

OpenAI ChatGPT was used to assist with manuscript organization, language refinement, and checking of numerical consistency. The authors retain responsibility for final verification, approval, and submission of the manuscript. No data or references were generated as substitutes for source material.

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